Pseudocode

Describing and Designing The Code

Today

- We'll explore what pseudocode is
- Learn how we write pseudocode
- Work through a basic example

What is pseudocode

- Language we use to write our solution algorithm
- Plain English
- “sue - doe - code”

Things computer programs do

- Receive data or information
- Output data or information
- Perform arithmetic
- Assign values to variables
- Compare values and make a decision
- Repeat processes

How To Write Pseudocode

- Each line of pseudocode revolves around an operation that occurs in the computer

Getting Data

- **read, get, prompt, input**
  - **read** is when we read from a file
  - **get** is when we get data from the keyboard
  - **prompt** is when we get it from the user by asking for it.
  - **input** is used when we have the data but we have to do something with it.

Storing (Assigning variables)

- Called "Assignment statements"
  - **initialize** (give it an initial value)
  - **set** assigns a value
  - **save** and **store** are for more permanent storage, like to a disk

Assignment

- **initialize** taxRate to 0.6
- **prompt** for subtotalAmount
- **set** tax to taxRate times subtotalAmount
  - **set** tax = taxRate * subtotalAmount

Be Specific

Bad

prompt for subtotalAmount

Good

prompt for subtotalAmount
with “What is the subtotal?”

Be Specific

Bad

display tax and total

Good

display “The tax is $“ + tax +
“and the total is $“ + total + “.”

Doing Math

- We can use words or symbols
  - **multiply** length times width to get area (words)
  - **area = length** * **width** (symbols)
- Symbols
  - + (addition)
  - - (subtraction)
  - * (multiplication)
  - / (division)
  - ( ) (parenthesis

Order of Operations

PEMDAS

Order of Operations

- Order of Operations in math applies in pseudocode and it applies in computer programming.
  - Operations in parenthesis are first
  - Then exponents
  - Then multiplication and division from left to right
  - Then addition and subtraction from left to right

Can you solve this?

5 * (3 + 4) - 2 * 8

Can you solve this?

5 * (3 + 4) - 2 * 8

5 * (7) - 2 * 8

Can you solve this?

5 * (3 + 4) - 2 * 8

5 * (7) - 2 * 8

35 - 16

19

It's important to get this right!

(3 + 4) * 5 is not the same as 3 + 4 * 5!!!

Handling Output

- **print, write, put, output,** or **display**
  - **print** is when the output goes to a printer.
  - **display** is when the output goes to the screen
  - **write** and **output** are when the output goes to a file or database.

Making Decisions

- IF, THEN, ELSE, ENDIF
  - IF is used to compare values
  - THEN is used to describe what you do when the comparison is true
  - ELSE is used to describe what you do when the comparison is false
  - ENDIF denotes the end of the decision

IF

```
IF the state is "Wisconsin" THEN
   set taxRate to 0.055
ENDIF
```

IF/ELSE

```
IF the state is "Wisconsin" THEN
  set taxRate to 0.055
ELSE
  set taxRate to 0
ENDIF
```

IF/ELSEIF/ELSE

```
IF the state is "Wisconsin" THEN
  set taxRate to 0.055
ELSEIF the state is "Minnesota" THEN
  set taxRate to 0.065
ELSE
  set taxRate to 0
ENDIF
```

Repeating

- WHILE and DO can be used when there's a sequence of processing steps to be repeated
  - WHILE describes the condition that makes us keep repeating the process
    - ENDWHILE designates the end of the process
- Sometimes you might see DO UNTIL instead.

WHILE

```
WHILE there are students to process
   read next name from roster
   display student name
ENDWHILE
```

Other kinds of loops

```
LOOP starting at 0 up to 10 in increments of 1
  DISPLAY "Hello World"
ENDLOOP
```

```
FOREACH person in people
  count = count + 1
END
```

A Pseudocode Example

initialize numberOfStudents to 0
initialize classGradesAsPercentages to 0
WHILE there are still students who need grades
    prompt for studentName
    prompt for gradeAsPercentage
    add one to numberOfStudents
    add gradeAsPercentage to classGradesAsPercentages
ENDWHILE
set average to
    classGradesAsPercentages divided by numberOfStudents
display average

A Pseudocode Example

initialize numberOfStudents to 0
initialize classGradesAsPercentages to 0
WHILE there are still students who need grades
    prompt for studentName with "What is the student name?"
    prompt for gradeAsPercentage with "What is the grade %"
    add one to numberOfStudents
    add gradeAsPercentage to classGradesAsPercentages
ENDWHILE
set average to
    classGradesAsPercentages divided by numberOfStudents
display "The average is " + average + "%"

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Keywords

- initialize
- set
- prompt
- display
- IF…THEN
  ELSE
  ENDIF
- IF..THEN
  ELSEIF
  ELSE
  END
- WHILE…DO
  ENDWHILE
- LOOP….
  ENDLOOP
- FOREACH
  ENDFOR

Simple Interest

- Calculate interest earned on an investment. Prompt for the principal, the rate, and the years. Display the interest accrued. Here's the formula:

- Interest = principal * rate * years

TEST PLANS!

- Variables: principal, rate, years
- Inputs: 5000, 4.8, 2
- Expected result: The interest is $480

Try writing the pseudocode.

- Calculate interest earned on an investment. Prompt for the principal, the rate, and the years. Display the interest accrued. Here's the formula:
- Interest = principal * rate * years

Pseudocode

CalculateInterest
  initialize interest to 0
  initialize output to “”
  initialize principle to 0
  initialize rate to 0
  initialize years to 0
  prompt for principal as “What is the principal?”
  prompt for rate as “What is the rate?”
  prompt for years as “How many years?”
  set interest = principal * (rate/100) * years
  set output = “The interest is $“ + interest
  display output
End

Other test plan?

- Variables: principal, rate, years
- Inputs: “abcdef”, 4.8%, 2
- Expected result: “Please enter numbers for the principal, rate, and years.”

Pseudocode

```
prompt for principal with "What is the principal?"
prompt for rate with "What is the rate?"
prompt for years with "How many years?"
IF rate is a number AND principal is a number
                    AND years is a number THEN
  set interest = principal * (rate/100) * years
  set output =  "The interest is $" + interest
ELSE
  set output =  "Please enter numbers for the principal, rate,
                 and years."
ENDIF

display output
```

Wrapping up

- Pseudocode is the written form of the **solution algorithm.**
- The style is flexible, but we have some standard keywords we'll use.
- It's a **design tool.**